IoT-Based Smart Home Automation System with Two-Way Control

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1. PROJECT OBJECTIVE

To design and implement a cost-effective, cloud-connected smart home system capable of controlling household appliances (lights and fans) via voice commands, mobile applications, and traditional physical wall switches.

2. HARDWARE ARCHITECTURE

- **Microcontroller:** ESP32 Development Board (handles Wi-Fi connectivity and logic processing).
- **Actuators:** 4-Channel 5V Relay Module (acts as an electronic bridge to switch 220V AC mains).
- **Input Interfaces:** Standard mechanical wall switches wired as low-voltage digital inputs.
- **Power Supply:** 5V USB adapter for the logic circuit.

3. SOFTWARE INTEGRATION

- **Firmware:** Written in C++ using the Arduino IDE.
- **IoT Platform:** Sinric Pro handles the remote device states and API bridges.
- **Voice Assistant:** Google Home application manages voice recognition, time schedules, and routine automation.

4. KEY ENGINEERING FEATURES

- **Two-Way Synchronization:** The system reads the physical switch positions and updates the cloud state in real-time. If a light is turned on manually, the mobile application immediately reflects the change, ensuring seamless operation even if the Wi-Fi connection drops.
- **Cloud Automations:** The system supports scheduled triggers and multi-device routines. A custom "Study Mode" voice command was programmed to simultaneously turn the desk lamp on and the room light off.
- **Safety and Isolation:** High-voltage AC mains are strictly isolated from the low-voltage logic circuits using optocoupled relays, ensuring a safe physical interface.

5. CONCLUSION & FUTURE SCOPE